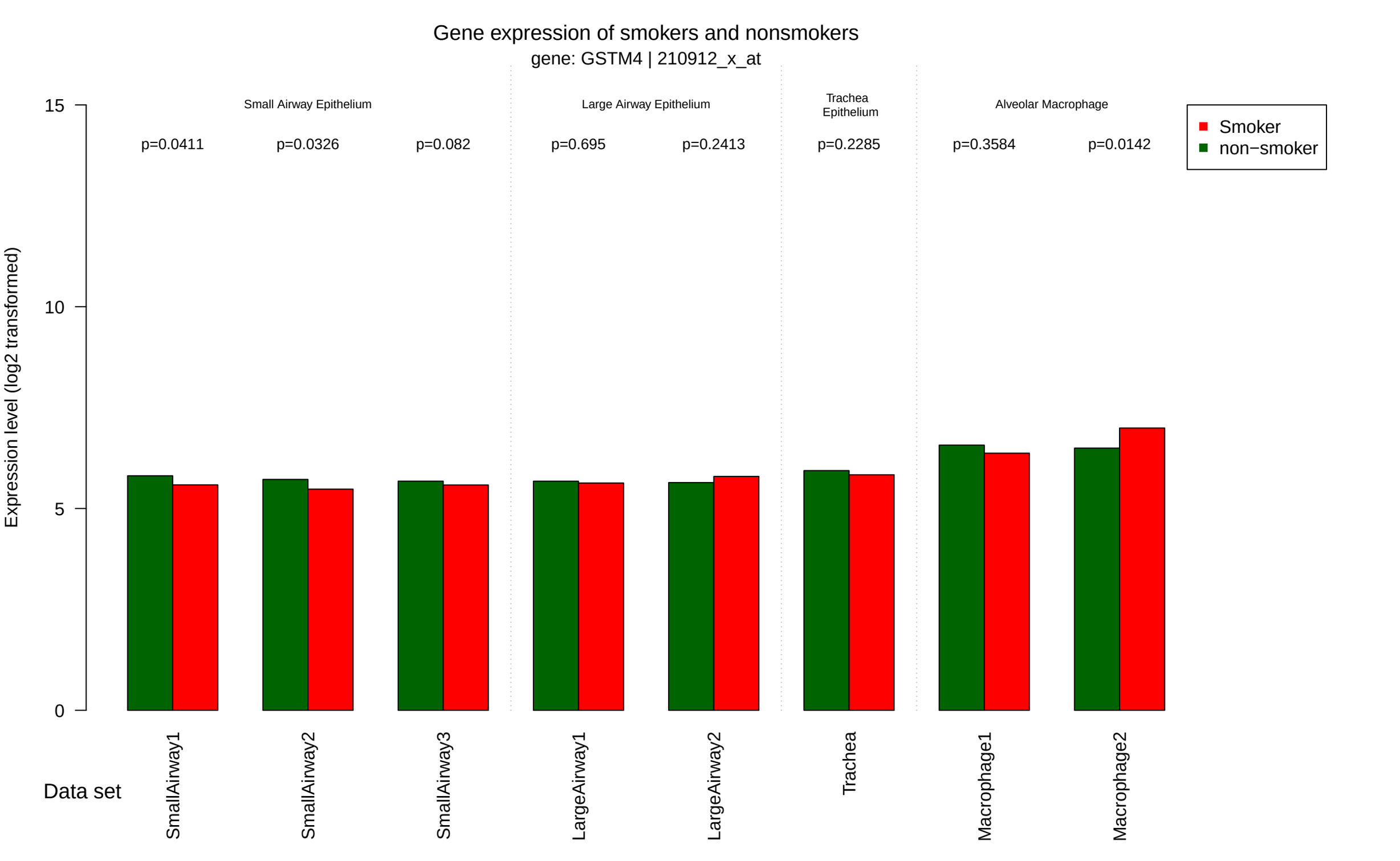
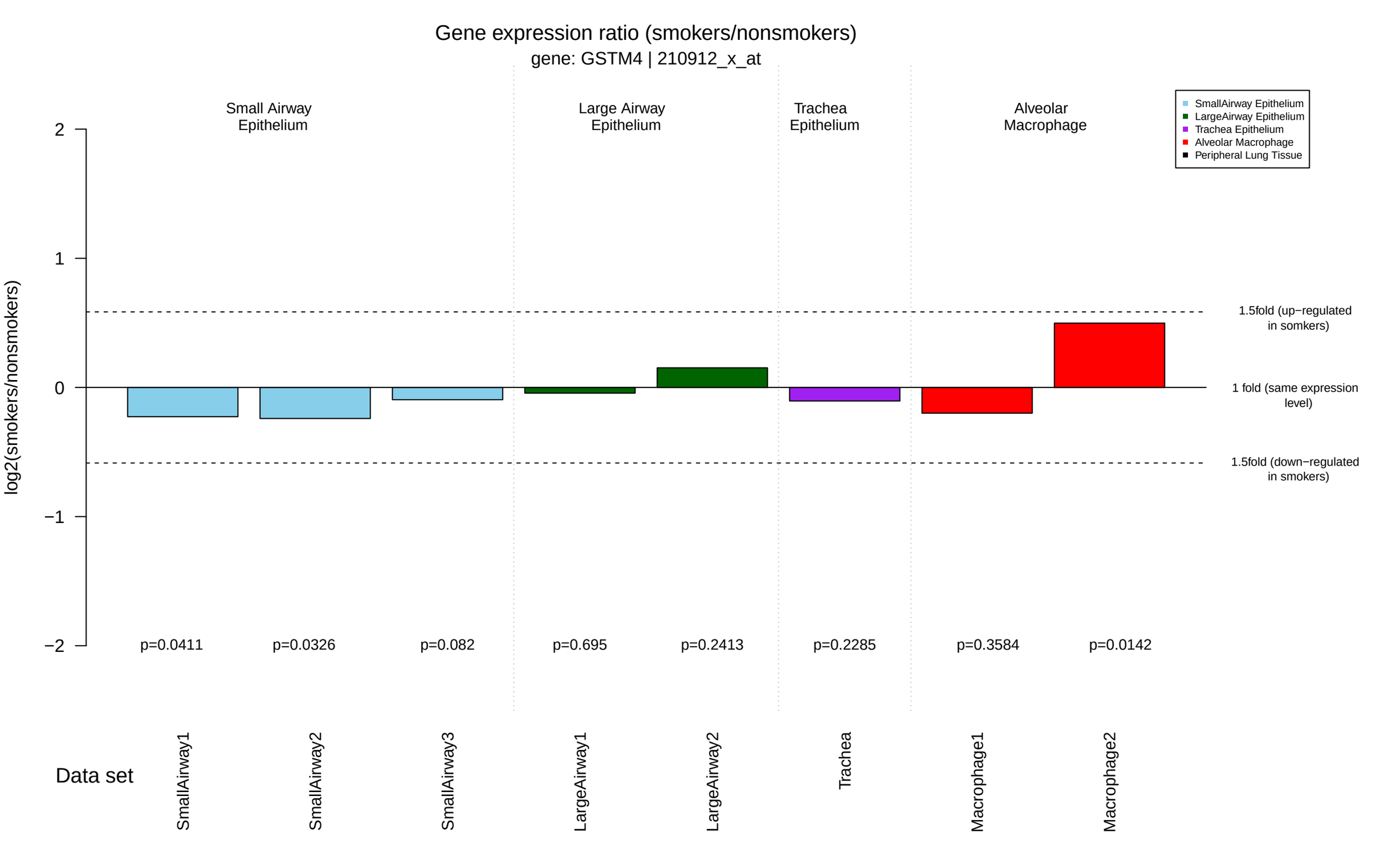
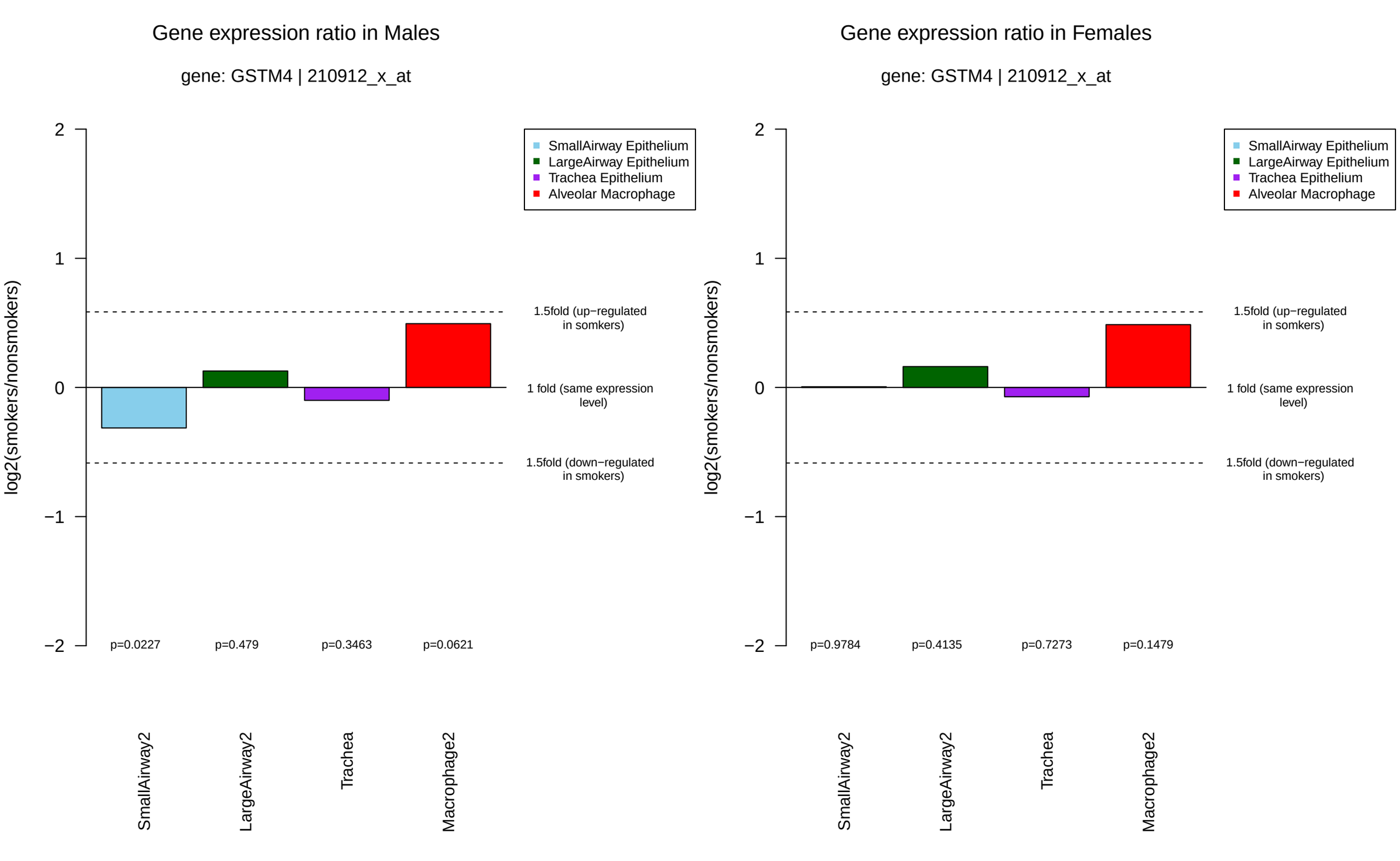
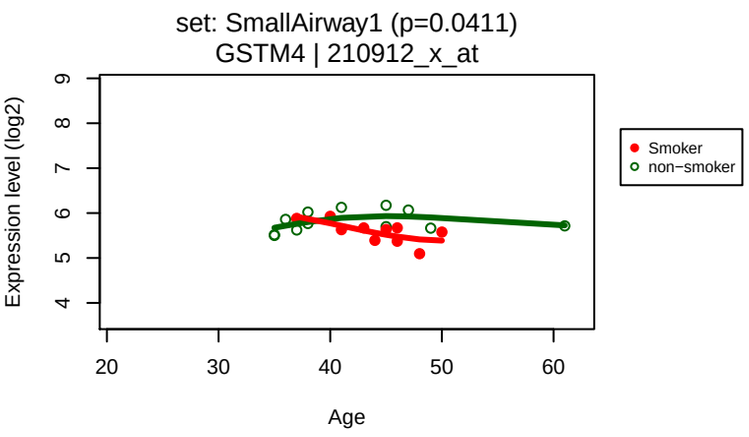
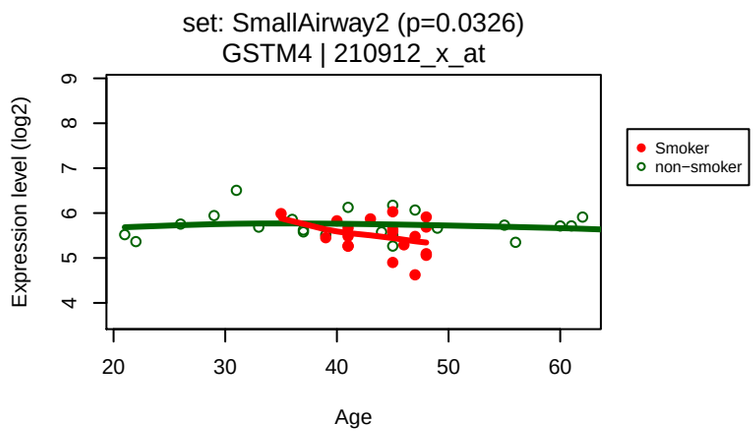
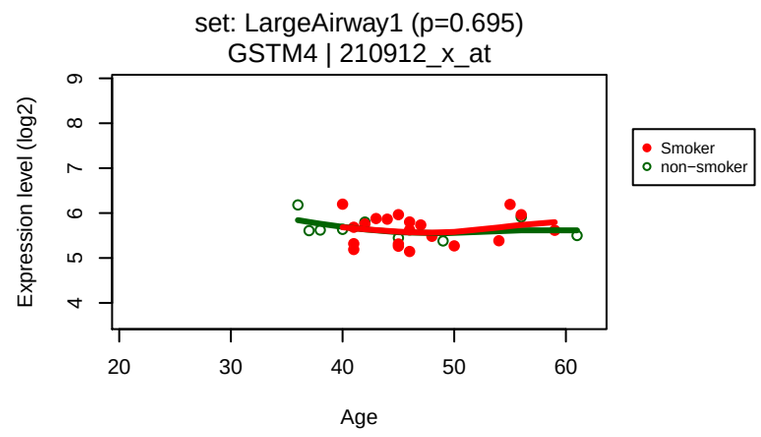

Gene expression of smokers and nonsmokers
gene: GSTM4 | 210912_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0411
p=0.0326
p=0.082
p=0.695
p=0.2413
p=0.2285
p=0.3584
p=0.0142
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

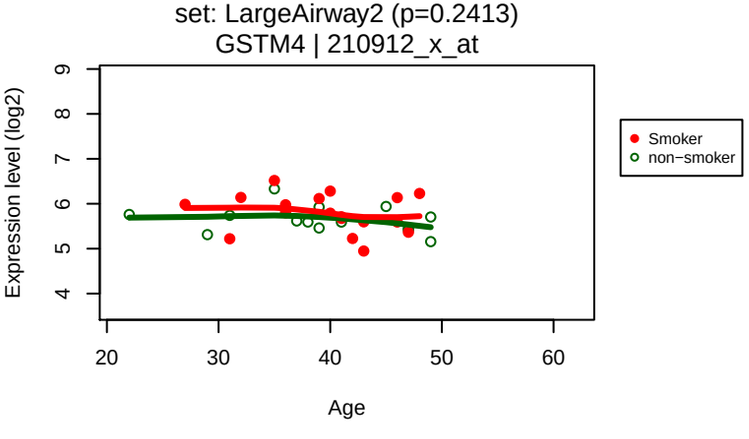

Gene expression ratio (smokers/nonsmokers)
gene: GSTM4 | 210912_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0411
p=0.0326
p=0.082
p=0.695
p=0.2413
p=0.2285
p=0.3584
p=0.0142
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

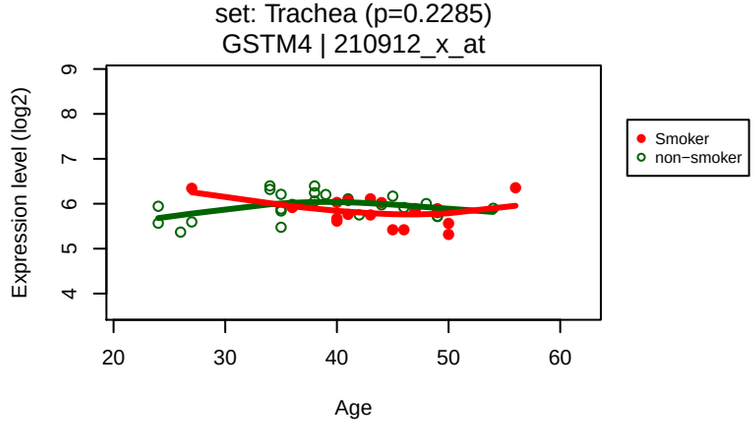

Gene expression ratio in Males
gene: GSTM4 | 210912_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0227
p=0.479
p=0.3463
p=0.0621
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GSTM4 | 210912_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9784
p=0.4135
p=0.7273
p=0.1479
SmallAirway2
LargeAirway2
Trachea
Macrophage2

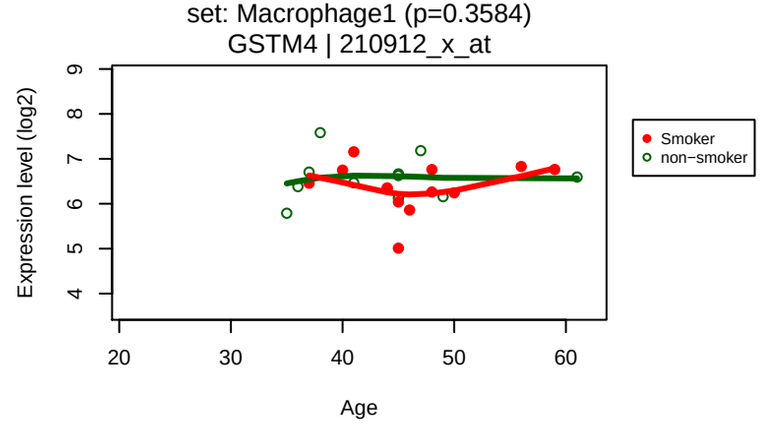

set: SmallAirway1 (p=0.0411)
GSTM4 | 210912_x_at
Expression level (log2)
Age
Smoker
non−smoker

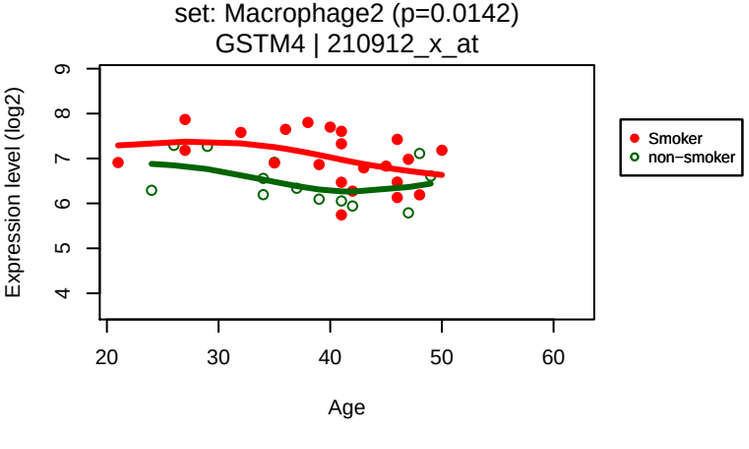

set: SmallAirway2 (p=0.0326)
GSTM4 | 210912_x_at
Expression level (log2)
Age
Smoker
non−smoker

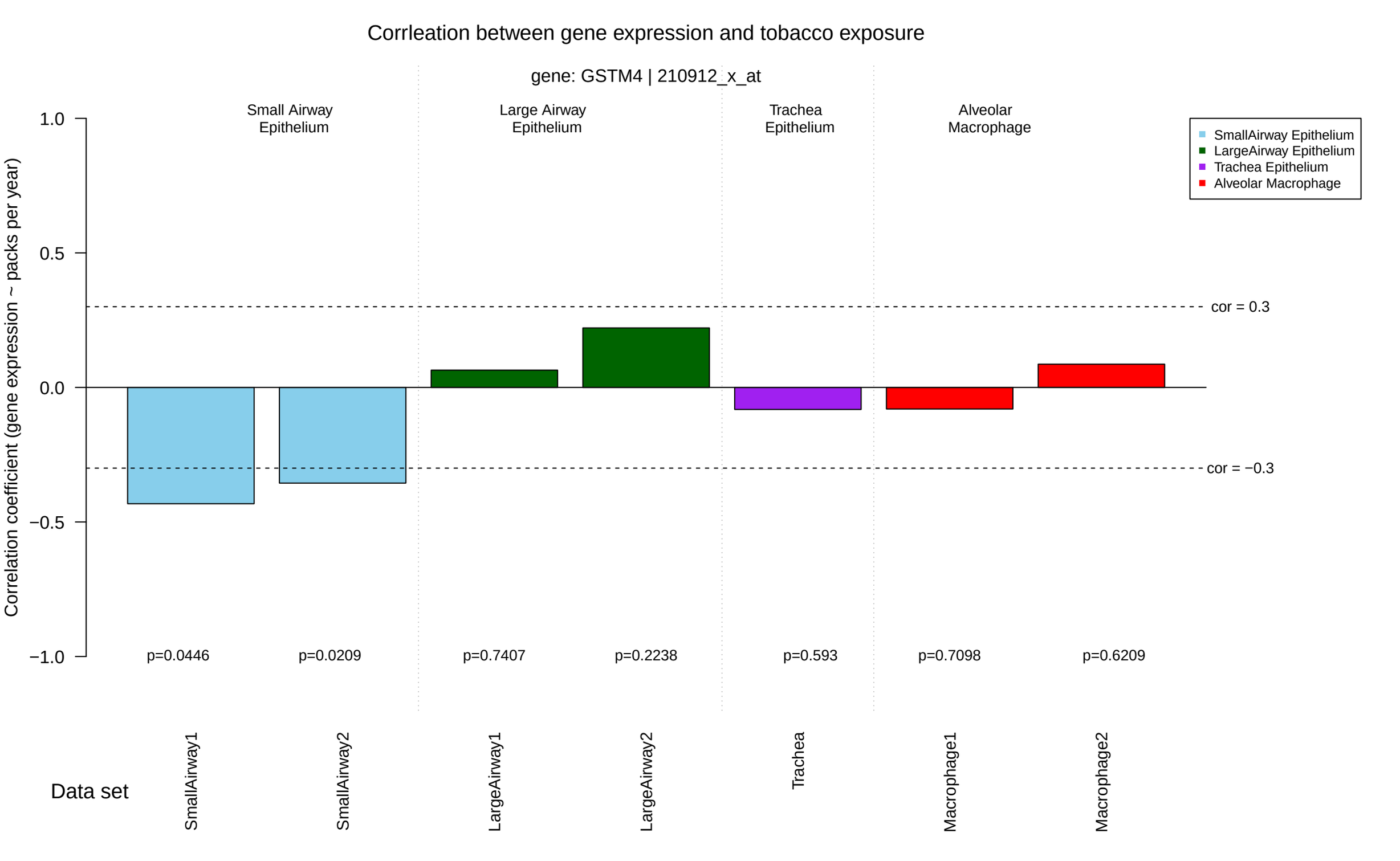

set: LargeAirway1 (p=0.695)
GSTM4 | 210912_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2413)
GSTM4 | 210912_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2285)
GSTM4 | 210912_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3584)
GSTM4 | 210912_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0142)
GSTM4 | 210912_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GSTM4 | 210912_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0446
p=0.0209
p=0.7407
p=0.2238
p=0.593
p=0.7098
p=0.6209
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2